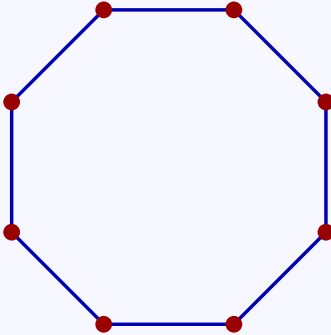


The Polygon Flip: Why 12 Lines Beat 8

Beam-only scaling says fewer lines are lighter. System effects reverse the tradeoff.

V6 Baseline
8-line octagon



Mass: 259 kg Do: 118 mm
Per-beam compression: $N/8$
8 knuckles $\times$ 0.11 kg = 0.88 kg
8 tethers $\times$ 67 m

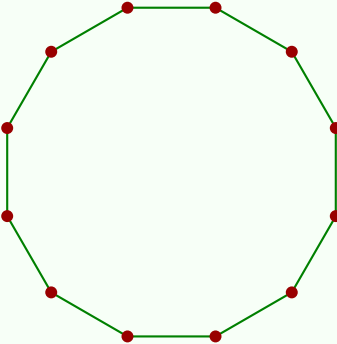
$\sin(\pi/8) = 0.383$ $n\sqrt{\sin(\pi/n)} = 2.86$

*Thick beams need thrust distribution:
expansion rotors at intermediate rings
to share the high per-beam load*

−71%

259 kg $\rightarrow$ 74 kg

V6.2 Optimum
12-line dodecagon



Mass: 74 kg Do: 95 mm
Per-beam compression: $N/12$ — 33% lower
12 knuckles $\times$ 0.10 kg = 1.20 kg
12 tethers $\times$ 67 m

$\sin(\pi/12) = 0.259$ $n\sqrt{\sin(\pi/n)} = 6.10$

*Thin beams handle compression locally:
hub-only expansion rotor ($n_{exp}=1$)
— a straight TRPT shaft —
no intermediate stations needed*

The tradeoff: Beam-only scaling favours fewer lines ($n\sqrt{\sin(\pi/n)} = 2.86$ at n=8 vs 6.10 at n=12). But system effects reverse this: thinner beams (95 vs 118 mm) enable smaller knuckles (0.10 vs 0.11 kg each), and per-beam compression drops 33%. The $2.19\times$ beam-mass penalty is offset by